

§14.5 The Chain Rule

Homework : 5,7,11,13,19,25,26,29,33,39,43,45,47,49,51

(1) The Chain Rule

$$z(t) = f(x(t), y(t)) \Rightarrow \frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} .$$

(由上節的 Differentials 或 Differentiable 的定義來感受上式.)

Example 1 : $z = f(x, y) = x^2 + xy + y^2$, $x = \cos t$, $y = \sin t$.

$$\begin{aligned} z &= f(x, y) = x^2 + xy + y^2 \\ \Rightarrow z &= \cos^2 t + \cos t \sin t + \sin^2 t \\ \frac{dz}{dt} &= 2 \cos t (-\sin t) + (-\sin t) \sin t + \cos t \cos t + 2 \sin t \cos t \\ &= (2 \cos t + \sin t)(-\sin t) + (\cos t + 2 \sin t) \cos t \\ &= (2x + y) \frac{dx}{dt} + (x + 2y) \frac{dy}{dt} . \end{aligned}$$

(2) Tree Diagram

$$z = (t, s) = f(x(t, s), y(t, s))$$

$$\begin{array}{c} \partial \quad z \quad \partial \\ \swarrow \quad \searrow \\ \partial \quad x \quad \partial \quad y \quad \partial \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ t \quad s \quad t \quad s \end{array} \Rightarrow \begin{cases} \frac{\partial z}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t} \\ \frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s} \end{cases}$$

(3) Implicit Differentiation

If $f(x, y, z) = 0$ and $z = g(x, y)$

$$\Rightarrow f(x, y, g(x, y)) = 0$$

$$\Rightarrow f_x + f_y \frac{dy}{dx} + f_z \frac{\partial z}{\partial x} = 0$$

$$\text{Since } \frac{dy}{dx} = 0 \Rightarrow \frac{\partial z}{\partial x} = -\frac{f_x}{f_z} . \quad \text{同理 } \frac{\partial z}{\partial y} = -\frac{f_y}{f_z} .$$

Example2 : $u = x^4 y + y^2 z^2$, $x = r s e^t$, $y = r s^2 e^{-t}$, $z = r^2 s \sin t$.

$$\text{Find } \left. \frac{\partial u}{\partial s} \right|_{\substack{r=2 \\ s=1 \\ t=0}} .$$

Solution :

$$\begin{aligned} \left. \frac{\partial u}{\partial s} \right|_{\substack{r=2 \\ s=1 \\ t=0}} &= 4x^3 y (r e^t) + (x^4 + 2y z^2) (2s r e^{-t}) + 2y^2 z (r^2 \sin t) \\ &= 128 + 64 \\ &= 192 . \end{aligned}$$

Example 3 : $x^3 + y^3 + z^3 + 6xyz = 1$. Find $\frac{\partial z}{\partial x}$.

Solution :

$$\begin{aligned} 3x^2 + 3z^2 \frac{\partial z}{\partial x} + 6yz + 6xy \frac{\partial z}{\partial x} &= 0 . \\ \Rightarrow \frac{\partial z}{\partial x} &= \frac{-x^2 - 2yz}{z^2 + 2xy} . \end{aligned}$$

Example 4 : Let f has continuous 2nd – order partial derivatives.

$$z = f(x, y) , \begin{cases} x = r^2 + s^2 \\ y = 2rs \end{cases} , \text{ find } \frac{\partial^2 z}{\partial r^2} .$$

Solution :

$$\begin{aligned} \frac{\partial z}{\partial r} &= f_x 2r + f_y 2s \\ \frac{\partial^2 z}{\partial r^2} &= (f_{xx} 2r + f_{xy} 2s) 2r + 2f_x + (f_{yx} 2r + f_{yy} 2s) 2s \\ &= 4r^2 f_{xx} + 8rs f_{xy} + 4s^2 f_{yy} + 2f_x \\ &\quad (f_{xy} = f_{yx}) \end{aligned}$$

Example 5 : $PV = 8.31T$. Find $\frac{dP}{dt}$ when $T = 300K$ and is increasing at a rate of $0.1 K/s$, and $V = 100L$ and is increasing at a rate of $0.2 L/s$.

Solution :

$$P = \frac{8.31T}{V} = \frac{8.31 \times 300}{100} = 24.93$$

$$\frac{dP}{dt}V + P\frac{dV}{dt} = 8.31\frac{dT}{dt} \quad (1)$$

$$\frac{dT}{dt} = 0.1, \quad \frac{dV}{dt} = 0.2 \quad T = 300, \quad V = 100, \quad P = 24.93$$

代入(1)式求 $\frac{dP}{dt}$.

Example 6 : $z = f(x, y)$, $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$

(i) Find $\frac{\partial^2 z}{\partial r^2}$, $\frac{\partial^2 z}{\partial \theta^2}$

(ii) Prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} + \frac{1}{r} \frac{\partial z}{\partial r}$.

Solution :

(i)

$$\frac{\partial z}{\partial r} = f_x \cos \theta + f_y \sin \theta$$

$$\frac{\partial^2 z}{\partial r^2} = (f_{xx} \cos \theta + f_{xy} \sin \theta) \cos \theta + (f_{yx} \cos \theta + f_{yy} \sin \theta) \sin \theta$$

$$= f_{xx} \cos^2 \theta + 2f_{xy} \sin \theta \cos \theta + f_{yy} \sin^2 \theta$$

$$\frac{\partial z}{\partial \theta} = f_x (-r \sin \theta) + f_y r \cos \theta$$

$$\frac{\partial^2 z}{\partial \theta^2} = [f_{xx} (-r \sin \theta) + f_{xy} r \cos \theta] (-r \sin \theta)$$

$$+ f_x (-r \cos \theta) + f_y (-r \sin \theta) + [f_{yx} (-r \sin \theta) + f_{yy} r \cos \theta] r \cos \theta.$$

$$= r^2 \sin^2 \theta f_{xx} - 2r \cos \theta \sin \theta f_{xy} + r^2 \cos^2 \theta f_{yy} - r \cos \theta f_x - r \sin \theta f_y.$$

(ii) 由 (i) 可得.

Example 7 : If $g(s, t) = f(s^2 - t^2, t^2 - s^2)$ and f is differentiable.

Show that g satisfies $t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0$.

Proof :

$$\frac{\partial g}{\partial s} = 2sf_x - 2sf_y$$

$$\frac{\partial g}{\partial t} = -2tf_x + 2tf_y$$

$$\Rightarrow t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0$$